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Explaining the Relative Nature of General and Special Relativity

Thomas DeGerlia[a]

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ABSTRACT: In this paper, we show that time dilation is not a property of a single system but a comparison between two systems, defined solely by their relative spatial distribution of mass. Specifically stated, the time dilation between two systems is governed by the relative DeGerlia Inertial Density D, defined as mass divided by mean radius about the compared axis. We first exemplify this comparative nature with one of the most fundamental equivalences in general relativity, the gravitational time-dilation equation, which computes a system's time dilation relative to flat space-time: $d\tau/dt = \sqrt{1 - r_s/R}$. We use this reference value to validate the calculation we perform using relative D alone. We then validate correct calculations for a diverse set of two-system examples, including transformed systems (similar to the gravitational time dilation above, where system 2 is a transformed version of system 1) as well as discrete system comparisons, across the classical and relativistic regimes. To conduct demonstration, we introduce the DeGerlia Time Dilation Framework (DTDF), a simple, mathematically equivalent framework fully defined by general relativity, offering a second frame from which to visualize the relative nature of time.

Keywords: time dilation, general relativity, special relativity, gravitational time dilation, Lorentz time dilation, DeGerlia inertial density, inertial density, moment of inertia, astrophysics, Schwarzschild radius, DTDF, DTD, degerlia time dilation

1 Introduction

General relativity[1] is a model for describing our universe that is fundamentally comparative: no quantity has physical meaning without a comparator. The pace of time is no exception: the time dilation between two systems is a product of the systems' relative geometric mass distributions. Directly from the gravitational time dilation formula, one can derive the underlying two-system logic, and ultimately, GTD is simply a comparison between a system's pace of time relative to the pace of time of a selected benchmark system, which in this case is a system of the same mass, but at its Schwarzschild radius. The formula requires only one mass and one radius, the second system being derived directly from the first. While this is the familiar representation, it obscures the comparative nature of relativity and implies that one need not compare two systems to characterize time. That implication is incorrect: there is no privileged perspective from which to characterize time. In this paper, we evaluate the physical property responsible for relativistic time dilation and explore the underlying formulation that quantifies the relative time dilation between two systems.

The relative flow of time between two systems is dictated by their relative DeGerlia inertial density D, and in this paper, we do a comparative analysis of relative time dilation across a variety of uniform spherical systems from classical to relativistic regimes, and show that the same D ratio, $k = D_1/D_2 = M_1R_2/(M_2R_1)$, yields the Schwarzschild time-dilation ratio exactly across every tested case. This establishes that the relative rate of time between systems is a direct consequence of their relative mass distributions. The result makes explicit a simple relation already contained in general relativity: given two systems' masses and radii, their relative time dilation is determined by their relative spatial distribu-

[a] DeGerlia Expert Consulting, 3000 Lawrence Street, Denver, CO, United States of America. E-mail: tom.degerlia@tomdegerlia.com

Table 1 System properties for each pair comparison case.

Property	Case 1 $M=M$, $\sim r_s$	Case 2 $M=M$, $\sim \frac{4}{3} r_s$	Case 3 $R=R$	Case 4 $\rho=\rho$	Case 5 $I=I$	Case 6 General	Case 7 Classical
M_1 (kg)	2.00000e+30	1.00000e+30	1.00000e+30	1.00000e+30	1.00000e+32	1.00000e+20	8.00000e+0
M_2 (kg)	2.00000e+30	1.00000e+30	1.00000e+24	1.00000e+27	1.00000e+30	1.00000e+30	1.00000e+0
R_1 (m)	2.97049e+3	1.98031e+3	1.00000e+8	1.00000e+8	1.00000e+7	2.00000e-7	2.00000e+0
R_2 (m)	7.00000e+8	1.00000e+8	1.00000e+8	1.00000e+7	1.00000e+8	1.00000e+7	1.00000e+0
ρ_1 (kg/m^3)	1.82161e+19	3.07406e+19	2.38732e+5	2.38732e+5	2.38732e+10	2.98416e+39	2.38732e-1
ρ_2 (kg/m^3)	1.39203e+3	2.38732e+5	2.38732e-1	2.38732e+5	2.38732e+5	2.38732e+8	2.38732e-1
I_1 (kg·m^2)	7.05907e+36	1.56865e+36	4.00000e+45	4.00000e+45	4.00000e+45	1.60000e+6	1.28000e+1
I_2 (kg·m^2)	3.92000e+47	4.00000e+45	4.00000e+39	4.00000e+40	4.00000e+45	4.00000e+43	4.00000e-1
D_1 (kg/m)	6.73289e+26	5.04972e+26	1.00000e+22	1.00000e+22	1.00000e+25	5.00000e+26	4.00000e+0
D_2 (kg/m)	2.85714e+21	1.00000e+22	1.00000e+16	1.00000e+20	1.00000e+22	1.00000e+23	1.00000e+0
r_{s1} (m)	2.97046e+3	1.48523e+3	1.48523e+3	1.48523e+3	1.48523e+5	1.48523e-7	1.18819e-26
r_{s2} (m)	2.97046e+3	1.48523e+3	1.48523e-3	1.48523e+0	1.48523e+3	1.48523e+3	1.48523e-27
$D_{\mathrm{norm},1}$	9.99991e-1	7.50001e-1	1.48523e-5	1.48523e-5	1.48523e-2	7.42617e-1	5.94093e-27
$D_{\mathrm{norm},2}$	4.24352e-6	1.48523e-5	1.48523e-11	1.48523e-7	1.48523e-5	1.48523e-4	1.48523e-27
k_s	9.99990e-1	7.50000e-1	1.48523e-5	1.48523e-5	1.48523e-2	7.42616e-1	5.94093e-27
k_d	9.99991e-1	7.50001e-1	1.48523e-5	1.48523e-5	1.48523e-2	7.42617e-1	5.94093e-27
GTD_{s1}	1-9.96838e-1	1-5.00000e-1	1-7.42619e-6	1-7.42619e-6	1-7.45394e-3	1-4.92670e-1	1-2.97046e-27
GTD_{s2}	1-2.12176e-6	1-7.42619e-6	1-7.42616e-12	1-7.42616e-8	1-7.42619e-6	1-7.42644e-5	1-7.42616e-28
GTD_{d1}	1-9.96948e-1	1-5.00001e-1	1-7.42619e-6	1-7.42619e-6	1-7.45395e-3	1-4.92670e-1	1-2.97047e-27
GTD_{d2}	1-2.12176e-6	1-7.42619e-6	1-7.42617e-12	1-7.42617e-8	1-7.42619e-6	1-7.42644e-5	1-7.42617e-28
DTD_1	9.99995e-1	8.66026e-1	3.85387e-3	3.85387e-3	1.21870e-1	8.61752e-1	7.70774e-14
DTD_2	2.05998e-3	3.85387e-3	3.85387e-6	3.85387e-4	3.85387e-3	1.21870e-2	3.85387e-14

Table 2 Derived ratios across cases.

Ratio	Case 1 $M=M$, $\sim r_s$	Case 2 $M=M$, $\sim \frac{4}{3} r_s$	Case 3 $R=R$	Case 4 $\rho=\rho$	Case 5 $I=I$	Case 6 General	Case 7 Classical
$\mathrm{DTD}_1/\mathrm{DTD}_2$	4.85439e+2	2.24716e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$\sqrt{D_1/D_2}$	4.85439e+2	2.24716e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$\sqrt{(M_1R_2)/(M_2R_1)}$	4.85439e+2	2.24716e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$\sqrt{(I_1/I_2)(R_2/R_1)^3}$	4.85439e+2	2.24716e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$(I_1/I_2)^{1/5}(\rho_1/\rho_2)^{3/10}$	4.85439e+2	2.24716e+2	1.00000e+3	1.00000e+1	3.16228e+1	7.07107e+1	2.00000e+0
$(M_1R_2)/(M_2R_1)$	2.35651e+5	5.04972e+4	1.00000e+6	1.00000e+2	1.00000e+3	5.00000e+3	4.00000e+0
D_1/D_2	2.35651e+5	5.04972e+4	1.00000e+6	1.00000e+2	1.00000e+3	5.00000e+3	4.00000e+0
$(I_1/I_2)(R_2/R_1)^3$	2.35651e+5	5.04972e+4	1.00000e+6	1.00000e+2	1.00000e+3	5.00000e+3	4.00000e+0
$\mathrm{GTD}_{s1}/\mathrm{GTD}_{s2}$	1-9.96838e-1	1-4.99996e-1	1-7.42618e-6	1-7.35193e-6	1-7.44657e-3	1-4.92632e-1	1-2.22785e-27
$\mathrm{GTD}_{d1}/\mathrm{GTD}_{d2}$	1-9.96948e-1	1-4.99997e-1	1-7.42619e-6	1-7.35193e-6	1-7.44658e-3	1-4.92633e-1	1-2.22785e-27
$\sqrt{1-\mathrm{DTD}_1^2}/\sqrt{1-\mathrm{DTD}_2^2}$	1-9.96948e-1	1-4.99997e-1	1-7.42619e-6	1-7.35193e-6	1-7.44658e-3	1-4.92633e-1	1-2.22785e-27
I_1/I_2	1.80078e-11	3.92163e-10	1.00000e+6	1.00000e+5	1.00000e+0	4.00000e-38	3.20000e+1
$(I_1/I_2)^{1/2}$	4.24356e-6	1.98031e-5	1.00000e+3	3.16228e+2	1.00000e+0	2.00000e-19	5.65685e+0
$(\rho_1/\rho_2)^{1/5}(R_1/R_2)$	7.09728e-3	1.31430e-2	1.58489e+1	1.00000e+1	1.00000e+0	3.31445e-8	2.00000e+0
$(I_1/I_2)^{1/5}$	7.09728e-3	1.31430e-2	1.58489e+1	1.00000e+1	1.00000e+0	3.31445e-8	2.00000e+0
M_1/M_2	1.00000e+0	1.00000e+0	1.00000e+6	1.00000e+3	1.00000e+2	1.00000e-10	8.00000e+0
$\sqrt{M_1/M_2}$	1.00000e+0	1.00000e+0	1.00000e+3	3.16228e+1	1.00000e+1	1.00000e-5	2.82843e+0
R_1/R_2	4.24356e-6	1.98031e-5	1.00000e+0	1.00000e+1	1.00000e-1	2.00000e-14	2.00000e+0
$\sqrt{R_1/R_2}$	2.05999e-3	4.45007e-3	1.00000e+0	3.16228e+0	3.16228e-1	1.41421e-7	1.41421e+0

tion of mass alone.

2 Experimental Model

The experimental model employed in this study comprises several components. The framework is derived directly from general relativity and is mathematically equivalent to established general-relativistic calculations. It neither relies upon nor introduces any principle not already established within the theories of general and special relativity; this paper leverages a framing of the existing theory of relativity for exploratory purposes.

We begin by introducing the DeGerlia Framework and then demonstrate its complete mathematical equivalence with general relativity. We then work through a series of examples, each demonstrating a different comparative case, and in each case, we validate the result using standard gravitational time-dilation calculations.

3 The DeGerlia Time Dilation Framework

3.1 What is the DeGerlia Time Dilation Framework?

First and foremost, the entirety of the principles described herein represents no deviation from general relativity. It represents a mathematically equivalent reframing useful in certain relativistic calculations. The author has simply asserted a complement to GTD, where the form of time dilation is equal to the square root of k, instead of the square root of $1-k$.

$$GTD = d\tau/dt = \sqrt{1-k}$$
$$DTD = \sqrt{k}$$
$$DTD^2 + GTD^2 = 1$$

While equivalent to GTD mathematically, the DTD form offers several advantages:

- It is mathematically very easy to use and visualize, requiring only the ratio of mass to radius in comparison to a static constant to perform gravitational time dilation calculations. Just to cite one of many examples: if I take the ratio of mass to radius for any system, that is, the ratio is to the 26th power, you are very close to black hole collapse. Greater than 26 = black hole. Less than 26, it isn't a black hole.
- One can convert between GTD and DTD at any time. GTD is always recoverable.
- It offers finite output over the entire range of input.
- It properly represents flat spacetime as an ideal state that can only be approached.
- It properly represents the Schwarzschild condition as the point where the gravitational escape velocity exceeds the speed of light, and an event horizon forms from our perspective. This is very attainable, however, the ideal state of zero time dilation is not.
- It properly reflects the singularity as an asymptotic approach to $r = 0$.

3.2 Definitions, Formulas and Constants Used Herein

Symbol	Definition / Value
c	Light Speed $299\,792\,458\,\mathrm{m\,s^{-1}}$
D	DeGerlia Inertial Density M/R_{axis}
D_{crit}	DeGerlia Threshold $c^2/2G = 6.732\,95\mathrm{e}26\,\mathrm{kg\,m^{-1}}$
DTD	DeGerlia Time Dilation DTD$=\sqrt{k}$
G	Gravitational Constant $6.674\,30\mathrm{e}{-11} \pm 1.5\mathrm{e}{-15}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$
GTD	Gravitational Time Dilation $d\tau/dt = \sqrt{1-k}$
k	Linear Scale Factor $v^2/c^2 = r_s/R = D/D_{crit}$
k_D	Linear Scale Factor (DeGerlia inertial density) D/D_{crit}
k_s	Linear Scale Factor (Schwarzschild) r_s/R
k_v	Linear Scale Factor (velocity) v^2/c^2
P	Inertial Density $I/V = D \cdot 3/10\pi$
r_s	Schwarzschild Radius $2GM/c^2$
STE	Space-time Equivalence [2] $k_D = M_1R_2/M_2R_1$

3.3 Inertia and Moment of Inertia

In classical physicsnertia dictates how a system's motion changes in response to an applied force. This applies to both linear and rotational motion. In rotational dynamics, the "moment of inertia"[3] I dictates how the angular motion of a system about a specific axis changes when a torque is applied, and is defined as the sum of each mass element multiplied by the square of its distance from the axis of rotation.

$$\text{Moment of Inertia: } I = \sum_i m_i r_i^2 \tag{1}$$

Using Equation (1), one can determine how a system's motion changes when energy is applied about an axis (torque). As it pertains to linear motion, inertia (I) is characterized solely by the system's mass.

3.4 General Relativity and Time Dilation

General Relativity provides the formula for calculating a system's time dilation relative to "flat space-time".

$$d\tau/dt = \sqrt{1-k} \tag{2}$$

where $k = r_s/R = 2GM/(Rc^2)$, M is the system's mass, and R its radius.

The equation requires only a single system's mass and radius, yet k is itself a ratio of two radii: the system's radius R and the Schwarzschild radius r_s of that same mass. The calculation is therefore a comparison of two systems: the system as given, and the same mass at its Schwarzschild radius.

3.5 Definition of a System

A "system" is defined as any discrete collection of particles and the space between them, or equivalently, any unit of space and the particles contained within it. The framework regards the universe not as a clean separation of discrete material systems separated by the vacuum of space, but as a gradient of material density pervasive throughout. The most extreme densities reside not within near-scale systems but within the cores of highly dense gravitational objects far from the human scale, such as black holes or subatomic particles, and form a gradient extending to the near-scale "vacuum of space" at which another gravitational source begins to dominate. The framework imposes no requirement for uniformity or symmetry, a clean boundary separating matter from vacuum, or any particular density profile. It requires only a mass and a radius about a chosen axis. A system may be specified, for example, as the Earth out to the mean radius of its surface. For a body such as Jupiter, which lacks a clear boundary separating surface from atmosphere, the system may instead be specified out to a chosen radius or density. For a large system, a spherical sample may be taken from within it to characterize the behavior of particular regions. There can be no system modeled under General Relativity that lacks either mass or radius.

3.6 System Time

For a whole system (as distinct from any subsystem), and from that whole system's perspective, the pace of time is static and all other systems are perceived relative to that benchmark. A subsystem possesses distinct physical properties and consequently exhibits behavior different from that of the system as a whole. In general relativity, time dilation scales precisely with the relative mass-to-radius ratio; a part of a system, therefore, cannot be expected to share the pace of time of the whole.

3.7 Inertial Density P

Inertial density P as shown in (4) is a fundamental system property[4] characterizing the distribution of inertia across any system, whether in the linear or rotational context, in any direction, or about any axis. If we were to redefine classical density as I/v (inertia/volume), because inertia is mass for one-dimensional motion, this would remain consistent with the classical definition of density, where $\rho = m/v$.

Inertial density may be characterized as the hollowness, or foaminess, of a system. Consider the configuration with the greatest moment of inertia for a given volume: a thin shell, in which all mass resides at the radius, displaced as far from the axis as the geometry permits and concentrated at the outer edge. A soap bubble exemplifies this configuration; per unit volume it possesses substantial inertia despite its low mass, representing the maximum-inertia structure for its volume. Conversely, a configuration of equal radius with its mass concentrated as an approximate point mass at the center represents the opposite limit: as the mass concentrates toward the center, the moment of inertia approaches zero. The uniform solid sphere lies between these limits. Subdividing such a sphere into smaller hollow spheres redistributes mass toward the periphery, so that the resulting foam-like structure retains a high moment of inertia for its volume.

3.8 Mathematics of Inertial Density

To derive the inertial density relationship, we begin with the established relationship between mass and density ρ as shown in (3):

$$\rho = \frac{m}{v} \tag{3}$$

where ρ is density, m is mass, and v is volume. Because mass is equivalent to linear inertia, and because density is m/v, we define inertial density P in terms of moment of inertia over volume:

$$\boxed{\text{Inertial Density} : P = \frac{I}{v}} \tag{4}$$

where inertial density P is equal to inertia I over volume v. Note, the moment of inertia I for a system may be distinct about every axis for systems that are not uniform[5]. For a uniform sphere[6], we substitute $I = \frac{2}{5}mr^2$ and $v = \frac{4}{3}\pi r^3$:

$$P = \frac{I}{v} = \frac{\frac{2}{5}mr^2}{\frac{4}{3}\pi r^3} = \frac{3m}{10\pi r} \tag{5}$$

Which can be simplified as follows:

$$P = \frac{m}{r} \times \frac{3}{10\pi} \tag{6}$$

$$P = \frac{m}{r} \times 9.549\,296\,585\text{e}{-2} \tag{7}$$

Therefore, for spherical systems, P can be calculated as the ratio of mass to radius, multiplied by the conversion factor $\frac{3}{10\pi}$ as shown in (7).

3.9 DeGerlia Inertial Density D

For all spherical systems, m/r multiplied by the geometric factor $3/(10\pi)$ yields the inertial density. ***DeGerlia inertial density*** is a simplified analog of inertial density I/v, equal to the mass-to-radius ratio m/r with the $3/(10\pi)$ geometric conversion factor omitted:

$$\boxed{\text{DeGerlia Inertial Density: } D = \frac{m}{r}} \tag{8}$$

DeGerlia inertial density (8) $D = m/r$ is related to inertial density P by the geometric factor $3/10\pi$ as shown in (9):

$$P = \frac{3}{10\pi} \times D \tag{9}$$

In a linear context, inertia is mass, and inertial density reduces directly to mass density $\rho = M/V$. From this perspective, inertial density is the rotational analog of mass density, paralleling the relationship between $F = ma$ and $\tau = I\alpha$. Equivalently, mass density is the linear interpretation of inertial density.

Just as ordinary density ($\rho = M/V$) is linear inertia (mass) distributed over volume, DeGerlia inertial density is rotational inertia distributed over volume. As stated in the prior work, *mass and density are to linear motion as moment of inertia and inertial density are to rotational motion.*

Because D is almost always used in a ratio that cancels all geometric constants, this metric is typically preferred over the geometrically accurate inertial density P for ease of use.

3.10 DeGerlia Threshold D_{crit}

The Schwarzschild condition can be expressed as a static universal constant threshold of DeGerlia inertial density D, rather than as a value that must be recalculated for every mass.

Setting $R = r_s$ (the Schwarzschild condition), the DeGerlia inertial density D becomes the threshold D_{crit}:

$$D_{crit} = \frac{m}{r_s} \tag{10}$$

Recall the Schwarzschild radius[7]:

$$r_s = \frac{2Gm}{c^2} \tag{11}$$

Substitute r_s into the expression for D_{crit}:

$$D_{crit} = \frac{m}{\frac{2Gm}{c^2}} = \frac{mc^2}{2Gm} = \frac{c^2}{2G} \tag{12}$$

Insert the fundamental constants[8], $c = 2.99792458\text{e}8\,\text{m}\,\text{s}^{-1}$ and $G = 6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}$:

$$\begin{aligned} D_{crit} &= \frac{(2.99792458\text{e}8)^2}{2 \times 6.67430\text{e}{-11}}\,\text{kg}\,\text{m}^{-1} \\ &= \frac{8.98755179\text{e}16}{1.334860\text{e}{-10}} \\ &= 6.73295\text{e}26\,\text{kg}\,\text{m}^{-1} \end{aligned}$$

$$\boxed{\text{DeGerlia Threshold: } D_{crit} = 6.73295\text{e}26\,\text{kg}\,\text{m}^{-1}} \tag{13}$$

This is the universal constant value of DeGerlia inertial density D at which the escape velocity equals the speed of light: the mass-over-radius threshold at which gravitational collapse (collapse of the neutron structure) occurs. Note this value is limited by the precision of the gravitational constant[9], $G = 6.67430\text{e}{-11} \pm 1.5\text{e}{-15}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}$. Just as the Schwarzschild radius, the DeGerlia threshold is universally applicable to spherical gravitational systems; examples include black holes or neutron stars, a spherical region of an interstellar cloud, or even a galactic cluster.

Rapid evaluation of the black hole condition. Because the threshold is a single constant, $D_{crit} \approx 6.73295\text{e}26\,\text{kg}\,\text{m}^{-1}$, the black hole condition may be estimated directly from the order of magnitude of $D = M/R$. Taking the difference of the exponents of M and R: a difference of 27 or greater guarantees the black hole condition; a difference of 25 or less precludes it; and a difference of exactly 26 is indeterminate and requires explicit calculation. For a system with $M = 1\text{e}30$ and $R = 1\text{e}10$, the exponent difference is $30 - 10 = 20$, well below 26, and the system does not satisfy the black hole condition.

3.11 DeGerlia Time Dilation

DeGerlia Time Dilation is the geometric complement of gravitational time dilation:

$$DTD = \sqrt{k} \tag{14}$$

where k is the scale factor of Equation 15. Whereas $GTD = \sqrt{1-k}$ is the surviving proper-time rate $d\tau/dt$, $DTD = \sqrt{k}$ is its complement.

3.12 The Linear Scale Factor k

The following identity relates the time-dilation ratio between two uniform spherical systems to their masses and radii.

$$k = \frac{I_1}{I_2}\left(\frac{R_2}{R_1}\right)^3 = \frac{D_1}{D_2} = \frac{M_1 R_2}{M_2 R_1} \tag{15}$$

where R_1, R_2 are the radii, M_1, M_2 are the masses, and $D_i = M_i/R_i$ is the DeGerlia inertial density of system i. The three forms in Equation 15 are algebraically identical; $(I_1/I_2)(R_2/R_1)^3$ exposes the inertia-radius structure, D_1/D_2 is the DeGerlia inertial density D ratio, and $M_1R_2/(M_2R_1)$ is the fully expanded form. A system's pace of time has meaning only relative to another pace of time (dilation is a comparative notion). The scale factor k governs the relative time dilation between two uniform spherical systems. The standard gravitational time-dilation calculation takes the Schwarzschild form $\sqrt{1-k}$, where $k = D/D_{crit} = r_s/R$. The D ratio $k = D_1/D_2$ and the Schwarzschild form give the same value. Via the Schwarzschild radius, the same scale factor is:

$$k = \frac{D}{D_{crit}} = \frac{r_s}{R} \tag{16}$$

where $D = M/R$ is the system's DeGerlia inertial density and $r_s = 2GM/c^2$ is its Schwarzschild radius. This is the k that appears in the standard $\sqrt{1-k}$ for GTD.

Time dilation is always:

$$\frac{d\tau}{dt} = \sqrt{1-k} \tag{17}$$

where $d\tau/dt$ is the time-dilation factor between the two systems and k is the DeGerlia inertial density ratio for the case being computed. The specific forms are:

- $k = D_1/D_2$ (Equation 15), via inertial density ratio
- $k = D/D_{crit} = r_s/R$ (Equation 16), via Schwarzschild radius

A given numerical value of k produces the same time-dilation result irrespective of the form in which it is expressed.

3.13 Properties of DTD

DTD is the geometric complement of gravitational time dilation and is part of the same general relativity. The translation between GTD and DTD is bound through the Pythagorean relationship between the two quantities.

- GTD and DTD are bound by the Pythagorean identity:

$$GTD^2 + DTD^2 = 1 \tag{18}$$

4 Relative Time Dilation Examples

Tables 1 and 2 present the complete system properties and derived ratios for seven system pairs spanning the cases that follow.

The first is the most familiar: the case in which both systems have the same mass. Gravitational time dilation is such a case, because to calculate it, one must compare a system with an identical system at its Schwarzschild radius ($M = M$, $R \approx r_s$); this demonstrates the two-system nature of all relativistic calculations, where the compared radius is the collapse-condition radius r_s. A second equal-mass calculation follows, comparing a large system to a system of equal mass at four-thirds the Schwarzschild radius ($M = M$, $R = \frac{4}{3}r_s$), a point known to be 50% time dilation for any system, for which $d\tau/dt = 0.5$. Two systems of equal radius are then treated ($R = R$), recovering the Lorentz factor described under Special Relativity. Calculations are also performed for systems of equal density ($\rho = \rho$) and equal moment of inertia ($I = I$), for a classical case comparing near-scale objects, and for a general unconstrained case, in which the relative pace of time may be determined between any two systems bearing no similarity to one another.

Across every case, the three algebraic forms of the scale factor k (Equation 15) produce identical values, and the time-dilation ratio computed via the DeGerlia route agrees with the gravitational time-dilation ratio computed independently from the standard Schwarzschild form, to the limit of precision imposed by G. No case requires a separate formula. The worked examples in the following subsections demonstrate, case by case, how each column of these tables is obtained.

4.1 Case 1: ($M_1 = M_2$, $R_1 \approx r_s$)

The two identical masses cancel, reducing the DeGerlia inertial density ratio to a ratio of radii.

$$k_D = \frac{M_1 R_2}{M_2 R_1 D_{crit}} = \frac{R_2}{R_1 D_{crit}}$$

Note that this reflects the k calculation for gravitational time dilation, where we compare with r_s. Because $GTD = 0$ exactly at r_s, we set the comparison radius to $1.00001\,r_s$, an "arbitrarily" small amount above r_s.

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	2.00000e30	2.00000e30
R (m)	2.97049e3	7.00000e8

Calculate DTD for each System:

$$DTD = \sqrt{k} \quad \text{where} \quad k = M/(RD_{crit})$$
$$k_1 = \frac{2.00000\text{e}30\,\text{kg}}{(2.97049\text{e}3\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 9.9999068441\text{e}{-1}$$
$$DTD_1 = \sqrt{9.9999068441\text{e}{-1}} = 9.9999534220\text{e}{-1}$$
$$k_2 = \frac{2.00000\text{e}30\,\text{kg}}{(7.00000\text{e}8\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 4.24352\text{e}{-6}$$
$$DTD_2 = \sqrt{4.24352\text{e}{-6}} = 2.05998\text{e}{-3}$$
$$\frac{DTD_1}{DTD_2} = \frac{9.9999534220\text{e}{-1}}{2.05998\text{e}{-3}} = 4.85439\text{e}2$$

Calculate the GTD ratio from DTD:

$$\frac{GTD_{d1}}{GTD_{d2}} = \frac{\sqrt{1-DTD_1^2}}{\sqrt{1-DTD_2^2}} = \frac{\sqrt{1-(9.9999534220\text{e}{-1})^2}}{\sqrt{1-(2.05998\text{e}{-3})^2}}$$
$$= \frac{\sqrt{1-9.9999068441\text{e}{-1}}}{\sqrt{1-4.24352\text{e}{-6}}}$$
$$= \frac{\sqrt{9.31559\text{e}{-6}}}{\sqrt{9.9999575648\text{e}{-1}}}$$
$$= \frac{3.05214\text{e}{-3}}{9.9999787824\text{e}{-1}} = 3.05215\text{e}{-3}$$
$$= 1 - 9.9694785\text{e}{-1}$$

Verify by calculating the GTD from the Schwarzschild Radius Ratio:

Schwarzschild radii:

$$r_{s1} = r_{s2} = \frac{2GM}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(2.00000\text{e}30\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 2.97046\text{e}3\,\text{m}$$

Calculate k values from the r_s/R ratio:

$$k_1 = \frac{r_{s1}}{R_1} = \frac{2.97046\text{e}3\,\text{m}}{2.97049\text{e}3\,\text{m}} = 9.9999000010\text{e}{-1}$$
$$k_2 = \frac{r_{s2}}{R_2} = \frac{2.97046\text{e}3\,\text{m}}{7.00000\text{e}8\,\text{m}} = 4.24352\text{e}{-6}$$

Calculate GTD via Schwarzschild radius ratio:

$$GTD_{s1} = \sqrt{1-k_1} = \sqrt{9.99990\text{e}{-6}} = 3.16226\text{e}{-3} = 1 - 9.9683774\text{e}{-1}$$
$$GTD_{s2} = \sqrt{1-k_2} = \sqrt{9.9999575648\text{e}{-1}} = 9.9999787824\text{e}{-1}$$
$$= 1 - 2.12176\text{e}{-6}$$
$$\frac{GTD_{s1}}{GTD_{s2}} = \frac{3.16226\text{e}{-3}}{9.9999787824\text{e}{-1}} = 3.16227\text{e}{-3}$$
$$= 1 - 9.9683773\text{e}{-1}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{GTD_{s1}}{GTD_{s2}} = 1 - 9.9683773\text{e}{-1} \quad (\text{via } r_s)$$
$$\frac{GTD_{d1}}{GTD_{d2}} = 1 - 9.9694785\text{e}{-1} \quad (\text{via } D_{crit})$$

4.2 Case 2: ($M_1 = M_2$, $R_1 = \frac{4}{3} r_s$)

The two identical masses cancel, reducing the DeGerlia inertial density ratio to a ratio of radii.

$$k_D = \frac{M_1 R_2}{M_2 R_1 D_{crit}} = \frac{R_2}{R_1 D_{crit}}$$

Note: This reflects the k calculation for gravitational time dilation. However, in this case, instead of comparing with r_s, we compare with $\frac{4}{3} r_s$, which corresponds to a gravitational time dilation of $GTD = 5.000\,00\text{e}{-1}$.

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	1.000 00e30	1.000 00e30
R (m)	1.980 31e3	1.000 00e8

Calculate DTD for each System:

$$DTD = \sqrt{k} \quad \text{where} \quad k = M/(RD_{crit})$$
$$k_1 = \frac{1.000\,00\text{e}30\,\text{kg}}{(1.980\,31\text{e}3\,\text{m})(6.732\,95\text{e}26\,\text{kg/m})} = 7.500\,01\text{e}{-1}$$
$$DTD_1 = \sqrt{7.500\,01\text{e}{-1}} = 8.660\,26\text{e}{-1}$$
$$k_2 = \frac{1.000\,00\text{e}30\,\text{kg}}{(1.000\,00\text{e}8\,\text{m})(6.732\,95\text{e}26\,\text{kg/m})} = 1.485\,23\text{e}{-5}$$
$$DTD_2 = \sqrt{1.485\,23\text{e}{-5}} = 3.853\,87\text{e}{-3}$$
$$\frac{DTD_1}{DTD_2} = \frac{8.660\,26\text{e}{-1}}{3.853\,87\text{e}{-3}} = 2.247\,16\text{e}2$$

Calculate the GTD ratio from DTD:

$$\frac{GTD_{d1}}{GTD_{d2}} = \frac{\sqrt{1 - DTD_1^2}}{\sqrt{1 - DTD_2^2}} = \frac{\sqrt{1 - (8.660\,26\text{e}{-1})^2}}{\sqrt{1 - (3.853\,87\text{e}{-3})^2}}$$
$$= \frac{\sqrt{1 - 7.500\,01\text{e}{-1}}}{\sqrt{1 - 1.485\,23\text{e}{-5}}}$$
$$= \frac{\sqrt{2.499\,99\text{e}{-1}}}{\sqrt{9.999\,851\,477\text{e}{-1}}}$$
$$= \frac{4.999\,99\text{e}{-1}}{9.999\,925\,738\,1\text{e}{-1}} = 5.000\,03\text{e}{-1}$$
$$= 1 - 4.999\,97\text{e}{-1}$$

Verify by calculating the GTD from the Schwarzschild Radius Ratio:

Schwarzschild radii:

$$r_{s1} = r_{s2} = \frac{2GM}{c^2} = \frac{2(6.674\,30\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.000\,00\text{e}30\,\text{kg})}{(2.997\,92\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.485\,23\text{e}3\,\text{m}$$

Calculate k values from the r_s/R ratio:

$$k_1 = \frac{r_{s1}}{R_1} = \frac{1.485\,23\text{e}3\,\text{m}}{1.980\,31\text{e}3\,\text{m}} = 7.500\,00\text{e}{-1}$$
$$k_2 = \frac{r_{s2}}{R_2} = \frac{1.485\,23\text{e}3\,\text{m}}{1.000\,00\text{e}8\,\text{m}} = 1.485\,23\text{e}{-5}$$

Calculate GTD via Schwarzschild radius ratio:

$$GTD_{s1} = \sqrt{1 - k_1} = \sqrt{2.500\,00\text{e}{-1}} = 5.000\,00\text{e}{-1} = 1 - 5.000\,00\text{e}{-1}$$
$$GTD_{s2} = \sqrt{1 - k_2} = \sqrt{9.999\,851\,477\text{e}{-1}} = 9.999\,925\,738\,1\text{e}{-1}$$
$$= 1 - 7.426\,19\text{e}{-6}$$
$$\frac{GTD_{s1}}{GTD_{s2}} = \frac{5.000\,00\text{e}{-1}}{9.999\,925\,738\,1\text{e}{-1}} = 5.000\,04\text{e}{-1}$$
$$= 1 - 4.999\,96\text{e}{-1}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{GTD_{s1}}{GTD_{s2}} = 1 - 4.999\,96\text{e}{-1} \quad (\text{via } r_s)$$
$$\frac{GTD_{d1}}{GTD_{d2}} = 1 - 4.999\,97\text{e}{-1} \quad (\text{via } D_{crit})$$

4.3 Case 3: ($R_1 = R_2$)

When the two systems share the same radius, the radius terms cancel in k_D:

$$k_D = \frac{M_1 R_2}{M_2 R_1 D_{crit}} = \frac{M_1}{M_2 D_{crit}}$$

The Lorentz factor [10] has the same form as Equation 17, with $k = v^2/c^2$. Since velocity is distance per unit time, v^2/c^2 is the square of a ratio of two lengths, with the shared time unit canceling.

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	1.000 00e30	1.000 00e24
R (m)	1.000 00e8	1.000 00e8

Calculate DTD for each System:

$$DTD = \sqrt{k} \quad \text{where} \quad k = M/(RD_{crit})$$
$$k_1 = \frac{1.000\,00\text{e}30\,\text{kg}}{(1.000\,00\text{e}8\,\text{m})(6.732\,95\text{e}26\,\text{kg/m})} = 1.485\,23\text{e}{-5}$$
$$DTD_1 = \sqrt{1.485\,23\text{e}{-5}} = 3.853\,87\text{e}{-3}$$
$$k_2 = \frac{1.000\,00\text{e}24\,\text{kg}}{(1.000\,00\text{e}8\,\text{m})(6.732\,95\text{e}26\,\text{kg/m})} = 1.485\,23\text{e}{-11}$$
$$DTD_2 = \sqrt{1.485\,23\text{e}{-11}} = 3.853\,87\text{e}{-6}$$
$$\frac{DTD_1}{DTD_2} = \frac{3.853\,87\text{e}{-3}}{3.853\,87\text{e}{-6}} = 1.000\,00\text{e}3$$

Calculate the GTD ratio from DTD:

$$\frac{GTD_{d1}}{GTD_{d2}} = \frac{\sqrt{1 - DTD_1^2}}{\sqrt{1 - DTD_2^2}} = \frac{\sqrt{1 - (3.853\,87\text{e}{-3})^2}}{\sqrt{1 - (3.853\,87\text{e}{-6})^2}}$$
$$= \frac{\sqrt{1 - 1.485\,23\text{e}{-5}}}{\sqrt{1 - 1.485\,23\text{e}{-11}}}$$
$$= \frac{\sqrt{9.999\,851\,477\text{e}{-1}}}{\sqrt{9.999\,999\,999\,851\,477\text{e}{-1}}}$$
$$= \frac{9.999\,925\,738\,1\text{e}{-1}}{9.999\,999\,999\,925\,738\,3\text{e}{-1}} = 9.999\,925\,738\,1\text{e}{-1}$$
$$= 1 - 7.426\,19\text{e}{-6}$$

Verify by calculating the GTD from the Schwarzschild Radius Ratio:

Schwarzschild radii:

$$r_{s1} = \frac{2GM_1}{c^2} = \frac{2(6.674\,30\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.000\,00\text{e}30\,\text{kg})}{(2.997\,92\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.485\,23\text{e}3\,\text{m}$$
$$r_{s2} = \frac{2GM_2}{c^2} = \frac{2(6.674\,30\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.000\,00\text{e}24\,\text{kg})}{(2.997\,92\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.485\,23\text{e}{-3}\,\text{m}$$

Calculate k values from the r_s/R ratio:

$$k_1 = \frac{r_{s1}}{R_1} = \frac{1.485\,23\text{e}3\,\text{m}}{1.000\,00\text{e}8\,\text{m}} = 1.485\,23\text{e}{-5}$$
$$k_2 = \frac{r_{s2}}{R_2} = \frac{1.485\,23\text{e}{-3}\,\text{m}}{1.000\,00\text{e}8\,\text{m}} = 1.485\,23\text{e}{-11}$$

Calculate GTD via Schwarzschild radius ratio:

$$GTD_{s1} = \sqrt{1 - k_1} = \sqrt{9.999\,851\,477\text{e}{-1}} = 9.999\,925\,738\,1\text{e}{-1}$$
$$= 1 - 7.426\,19\text{e}{-6}$$
$$GTD_{s2} = \sqrt{1 - k_2} = \sqrt{9.999\,999\,999\,851\,477\text{e}{-1}}$$
$$= 9.999\,999\,999\,925\,738\,4\text{e}{-1} = 1 - 7.426\,16\text{e}{-12}$$
$$\frac{GTD_{s1}}{GTD_{s2}} = \frac{9.999\,925\,738\,1\text{e}{-1}}{9.999\,999\,999\,925\,738\,4\text{e}{-1}}$$
$$= 9.999\,925\,738\,2\text{e}{-1} = 1 - 7.426\,18\text{e}{-6}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{GTD_{s1}}{GTD_{s2}} = 1 - 7.426\,18\text{e}{-6} \quad (\text{via } r_s)$$
$$\frac{GTD_{d1}}{GTD_{d2}} = 1 - 7.426\,19\text{e}{-6} \quad (\text{via } D_{crit})$$

4.4 Case 4: $(\rho_1 = \rho_2)$

When the two systems share the same density, $M \propto R^3$, so $M_1/M_2 = (R_1/R_2)^3$. Substituting into k_D:

$$k_D = \frac{M_1 R_2}{M_2 R_1 D_{crit}} = \left(\frac{R_1}{R_2}\right)^3 \cdot \frac{R_2}{R_1 D_{crit}} = \frac{1}{D_{crit}} \left(\frac{R_1}{R_2}\right)^2$$

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	1.00000e30	1.00000e27
R (m)	1.00000e8	1.00000e7

Calculate DTD for each System:

$$DTD = \sqrt{k} \quad \text{where} \quad k = M/(RD_{crit})$$
$$k_1 = \frac{1.00000\text{e}30\,\text{kg}}{(1.00000\text{e}8\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 1.48523\text{e}{-5}$$
$$DTD_1 = \sqrt{1.48523\text{e}{-5}} = 3.85387\text{e}{-3}$$
$$k_2 = \frac{1.00000\text{e}27\,\text{kg}}{(1.00000\text{e}7\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 1.48523\text{e}{-7}$$
$$DTD_2 = \sqrt{1.48523\text{e}{-7}} = 3.85387\text{e}{-4}$$
$$\frac{DTD_1}{DTD_2} = \frac{3.85387\text{e}{-3}}{3.85387\text{e}{-4}} = 1.00000\text{e}1$$

Calculate the GTD ratio from DTD:

$$\frac{GTD_{d1}}{GTD_{d2}} = \frac{\sqrt{1-DTD_1^2}}{\sqrt{1-DTD_2^2}} = \frac{\sqrt{1-(3.85387\text{e}{-3})^2}}{\sqrt{1-(3.85387\text{e}{-4})^2}}$$
$$= \frac{\sqrt{1-1.48523\text{e}{-5}}}{\sqrt{1-1.48523\text{e}{-7}}}$$
$$= \frac{\sqrt{9.999851477\text{e}{-1}}}{\sqrt{9.99999851477\text{e}{-1}}}$$
$$= \frac{9.9999257381\text{e}{-1}}{9.999999257383\text{e}{-1}} = 9.9999264807\text{e}{-1}$$
$$= 1 - 7.35193\text{e}{-6}$$

Verify by calculating the GTD from the Schwarzschild Radius Ratio:

Schwarzschild radii:

$$r_{s1} = \frac{2GM_1}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\text{e}30\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.48523\text{e}3\,\text{m}$$
$$r_{s2} = \frac{2GM_2}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\text{e}27\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.48523\,\text{m}$$

Calculate k values from the r_s/R ratio:

$$k_1 = \frac{r_{s1}}{R_1} = \frac{1.48523\text{e}3\,\text{m}}{1.00000\text{e}8\,\text{m}} = 1.48523\text{e}{-5}$$
$$k_2 = \frac{r_{s2}}{R_2} = \frac{1.48523\,\text{m}}{1.00000\text{e}7\,\text{m}} = 1.48523\text{e}{-7}$$

Calculate GTD via Schwarzschild radius ratio:

$$GTD_{s1} = \sqrt{1-k_1} = \sqrt{9.999851477\text{e}{-1}} = 9.9999257381\text{e}{-1}$$
$$= 1 - 7.42619\text{e}{-6}$$
$$GTD_{s2} = \sqrt{1-k_2} = \sqrt{9.99999851477\text{e}{-1}} = 9.999999257384\text{e}{-1}$$
$$= 1 - 7.42616\text{e}{-8}$$
$$\frac{GTD_{s1}}{GTD_{s2}} = \frac{9.9999257381\text{e}{-1}}{9.999999257384\text{e}{-1}}$$
$$= 9.9999264807\text{e}{-1} = 1 - 7.35193\text{e}{-6}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{GTD_{s1}}{GTD_{s2}} = 1 - 7.35193\text{e}{-6} \quad (\text{via } r_s)$$
$$\frac{GTD_{d1}}{GTD_{d2}} = 1 - 7.35193\text{e}{-6} \quad (\text{via } D_{crit})$$

4.5 Case 5: $(I_1 = I_2)$

When the two systems share the same moment of inertia, $M_1 R_1^2 = M_2 R_2^2$, so $M_1/M_2 = (R_2/R_1)^2$. Substituting into k_D:

$$k_D = \frac{M_1 R_2}{M_2 R_1 D_{crit}} = \left(\frac{R_2}{R_1}\right)^2 \cdot \frac{R_2}{R_1 D_{crit}} = \frac{1}{D_{crit}} \left(\frac{R_2}{R_1}\right)^3$$

The DeGerlia inertial density ratio reduces to the cube of the radius ratio.

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	1.00000e32	1.00000e30
R (m)	1.00000e7	1.00000e8

Calculate DTD for each System:

$$DTD = \sqrt{k} \quad \text{where} \quad k = M/(RD_{crit})$$
$$k_1 = \frac{1.00000\text{e}32\,\text{kg}}{(1.00000\text{e}7\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 1.48523\text{e}{-2}$$
$$DTD_1 = \sqrt{1.48523\text{e}{-2}} = 1.21870\text{e}{-1}$$
$$k_2 = \frac{1.00000\text{e}30\,\text{kg}}{(1.00000\text{e}8\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 1.48523\text{e}{-5}$$
$$DTD_2 = \sqrt{1.48523\text{e}{-5}} = 3.85387\text{e}{-3}$$
$$\frac{DTD_1}{DTD_2} = \frac{1.21870\text{e}{-1}}{3.85387\text{e}{-3}} = 3.16228\text{e}1$$

Calculate the GTD ratio from DTD:

$$\frac{GTD_{d1}}{GTD_{d2}} = \frac{\sqrt{1-DTD_1^2}}{\sqrt{1-DTD_2^2}} = \frac{\sqrt{1-(1.21870\text{e}{-1})^2}}{\sqrt{1-(3.85387\text{e}{-3})^2}}$$
$$= \frac{\sqrt{1-1.48523\text{e}{-2}}}{\sqrt{1-1.48523\text{e}{-5}}}$$
$$= \frac{\sqrt{9.851477\text{e}{-1}}}{\sqrt{9.999851477\text{e}{-1}}}$$
$$= \frac{9.9254605\text{e}{-1}}{9.9999257381\text{e}{-1}} = 9.9255342\text{e}{-1}$$
$$= 1 - 7.44658\text{e}{-3}$$

Verify by calculating the GTD from the Schwarzschild Radius Ratio:

Schwarzschild radii:

$$r_{s1} = \frac{2GM_1}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\text{e}32\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.48523\text{e}5\,\text{m}$$
$$r_{s2} = \frac{2GM_2}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\text{e}30\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.48523\text{e}3\,\text{m}$$

Calculate k values from the r_s/R ratio:

$$k_1 = \frac{r_{s1}}{R_1} = \frac{1.48523\text{e}5\,\text{m}}{1.00000\text{e}7\,\text{m}} = 1.48523\text{e}{-2}$$
$$k_2 = \frac{r_{s2}}{R_2} = \frac{1.48523\text{e}3\,\text{m}}{1.00000\text{e}8\,\text{m}} = 1.48523\text{e}{-5}$$

Calculate GTD via Schwarzschild radius ratio:

$$GTD_{s1} = \sqrt{1-k_1} = \sqrt{9.851477\text{e}{-1}} = 9.9254606\text{e}{-1} = 1 - 7.45394\text{e}{-3}$$
$$GTD_{s2} = \sqrt{1-k_2} = \sqrt{9.999851477\text{e}{-1}} = 9.9999257381\text{e}{-1}$$
$$= 1 - 7.42619\text{e}{-6}$$
$$\frac{GTD_{s1}}{GTD_{s2}} = \frac{9.9254606\text{e}{-1}}{9.9999257381\text{e}{-1}} = 9.9255343\text{e}{-1}$$
$$= 1 - 7.44657\text{e}{-3}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{GTD_{s1}}{GTD_{s2}} = 1 - 7.44657\text{e}{-3} \quad (\text{via } r_s)$$
$$\frac{GTD_{d1}}{GTD_{d2}} = 1 - 7.44658\text{e}{-3} \quad (\text{via } D_{crit})$$

4.6 Case 6: (General, unconstrained)

When no constraint is applied between the systems, k_D retains its full form:

$$k_D = \frac{M_1 R_2}{M_2 R_1 D_{crit}}$$

Both mass and radius must be specified independently for each system.

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	1.00000e20	1.00000e30
R (m)	2.00000e−7	1.00000e7

Calculate DTD for each System:

$$DTD = \sqrt{k} \quad \text{where} \quad k = M/(RD_{crit})$$
$$k_1 = \frac{1.00000\text{e}20\,\text{kg}}{(2.00000\text{e}{-7}\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 7.42617\text{e}{-1}$$
$$DTD_1 = \sqrt{7.42617\text{e}{-1}} = 8.61752\text{e}{-1}$$
$$k_2 = \frac{1.00000\text{e}30\,\text{kg}}{(1.00000\text{e}7\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 1.48523\text{e}{-4}$$
$$DTD_2 = \sqrt{1.48523\text{e}{-4}} = 1.21870\text{e}{-2}$$
$$\frac{DTD_1}{DTD_2} = \frac{8.61752\text{e}{-1}}{1.21870\text{e}{-2}} = 7.07107\text{e}1$$

Calculate the GTD ratio from DTD:

$$\frac{GTD_{d1}}{GTD_{d2}} = \frac{\sqrt{1-DTD_1^2}}{\sqrt{1-DTD_2^2}} = \frac{\sqrt{1-(8.61752\text{e}{-1})^2}}{\sqrt{1-(1.21870\text{e}{-2})^2}}$$
$$= \frac{\sqrt{1-7.42617\text{e}{-1}}}{\sqrt{1-1.48523\text{e}{-4}}}$$
$$= \frac{\sqrt{2.57383\text{e}{-1}}}{\sqrt{9.99851477\text{e}{-1}}}$$
$$= \frac{5.07330\text{e}{-1}}{9.999257356\text{e}{-1}} = 5.07367\text{e}{-1}$$
$$= 1 - 4.92633\text{e}{-1}$$

Verify by calculating the GTD from the Schwarzschild Radius Ratio:
Schwarzschild radii:

$$r_{s1} = \frac{2GM_1}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\text{e}20\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.48523\text{e}{-7}\,\text{m}$$
$$r_{s2} = \frac{2GM_2}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\text{e}30\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.48523\text{e}3\,\text{m}$$

Calculate k values from the r_s/R ratio:

$$k_1 = \frac{r_{s1}}{R_1} = \frac{1.48523\text{e}{-7}\,\text{m}}{2.00000\text{e}{-7}\,\text{m}} = 7.42616\text{e}{-1}$$
$$k_2 = \frac{r_{s2}}{R_2} = \frac{1.48523\text{e}3\,\text{m}}{1.00000\text{e}7\,\text{m}} = 1.48523\text{e}{-4}$$

Calculate GTD via Schwarzschild radius ratio:

$$GTD_{s1} = \sqrt{1-k_1} = \sqrt{2.57384\text{e}{-1}} = 5.07330\text{e}{-1} = 1 - 4.92670\text{e}{-1}$$
$$GTD_{s2} = \sqrt{1-k_2} = \sqrt{9.99851477\text{e}{-1}} = 9.999257356\text{e}{-1}$$
$$= 1 - 7.42644\text{e}{-5}$$
$$\frac{GTD_{s1}}{GTD_{s2}} = \frac{5.07330\text{e}{-1}}{9.999257356\text{e}{-1}} = 5.07368\text{e}{-1}$$
$$= 1 - 4.92632\text{e}{-1}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{GTD_{s1}}{GTD_{s2}} = 1 - 4.92632\text{e}{-1} \quad (\text{via } r_s)$$
$$\frac{GTD_{d1}}{GTD_{d2}} = 1 - 4.92633\text{e}{-1} \quad (\text{via } D_{crit})$$

4.7 Case 7: (Classical)

This is a static-density, isometric case that follows the same model as Case 4.

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	8.00000	1.00000
R (m)	2.00000	1.00000

Calculate DTD for each System:

$$DTD = \sqrt{k} \quad \text{where} \quad k = M/(RD_{crit})$$
$$k_1 = \frac{8.00000\,\text{kg}}{(2.00000\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 5.94093\text{e}{-27}$$
$$DTD_1 = \sqrt{5.94093\text{e}{-27}} = 7.70774\text{e}{-14}$$
$$k_2 = \frac{1.00000\,\text{kg}}{(1.00000\,\text{m})(6.73295\text{e}26\,\text{kg/m})} = 1.48523\text{e}{-27}$$
$$DTD_2 = \sqrt{1.48523\text{e}{-27}} = 3.85387\text{e}{-14}$$
$$\frac{DTD_1}{DTD_2} = \frac{7.70774\text{e}{-14}}{3.85387\text{e}{-14}} = 2.00000$$

Calculate the GTD ratio from DTD:

$$\frac{GTD_{d1}}{GTD_{d2}} = \frac{\sqrt{1-DTD_1^2}}{\sqrt{1-DTD_2^2}} = \frac{\sqrt{1-(7.70774\text{e}{-14})^2}}{\sqrt{1-(3.85387\text{e}{-14})^2}}$$
$$= \frac{\sqrt{1-5.94093\text{e}{-27}}}{\sqrt{1-1.48523\text{e}{-27}}}$$
$$= \frac{\sqrt{9.9999999999999999999999999940590\text{7e}{-1}}}{\sqrt{9.9999999999999999999999999985147\text{7e}{-1}}}$$
$$= \frac{9.9999999999999999999999999970295\text{3e}{-1}}{9.999999999999999999999999999257383\text{e}{-1}}$$
$$= 9.9999999999999999999999999977721\text{5e}{-1}$$
$$= 1 - 2.22785\text{e}{-27}$$

Verify by calculating the GTD from the Schwarzschild Radius Ratio:
Schwarzschild radii:

$$r_{s1} = \frac{2GM_1}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(8.00000\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.18819\text{e}{-26}\,\text{m}$$
$$r_{s2} = \frac{2GM_2}{c^2} = \frac{2(6.67430\text{e}{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2})(1.00000\,\text{kg})}{(2.99792\text{e}8\,\text{m}\,\text{s}^{-1})^2}$$
$$= 1.48523\text{e}{-27}\,\text{m}$$

Calculate k values from the r_s/R ratio:

$$k_1 = \frac{r_{s1}}{R_1} = \frac{1.18819\text{e}{-26}\,\text{m}}{2.00000\,\text{m}} = 5.94093\text{e}{-27}$$
$$k_2 = \frac{r_{s2}}{R_2} = \frac{1.48523\text{e}{-27}\,\text{m}}{1.00000\,\text{m}} = 1.48523\text{e}{-27}$$

Calculate GTD via Schwarzschild radius ratio:

$$GTD_{s1} = \sqrt{1-k_1}$$
$$= \sqrt{9.9999999999999999999999999940590\text{7e}{-1}}$$
$$= 9.9999999999999999999999999970295\text{4e}{-1}$$
$$= 1 - 2.97046\text{e}{-27}$$
$$GTD_{s2} = \sqrt{1-k_2}$$
$$= \sqrt{9.9999999999999999999999999985147\text{7e}{-1}}$$
$$= 9.999999999999999999999999999257384\text{e}{-1}$$
$$= 1 - 7.42616\text{e}{-28}$$
$$\frac{GTD_{s1}}{GTD_{s2}} = \frac{9.9999999999999999999999999970295\text{4e}{-1}}{9.999999999999999999999999999257384\text{e}{-1}}$$
$$= 9.9999999999999999999999999977721\text{5e}{-1}$$
$$= 1 - 2.22785\text{e}{-27}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{GTD_{s1}}{GTD_{s2}} = 1 - 2.22785\text{e}{-27} \quad (\text{via } r_s)$$
$$\frac{GTD_{d1}}{GTD_{d2}} = 1 - 2.22785\text{e}{-27} \quad (\text{via } D_{crit})$$

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Conflicts of Interest

The author declares no competing interests.

Data Availability

The complete source materials for this paper, including all LaTeX source files, derivation documents, and revision history, are publicly available at: `https://github.com/denverdata/academic_InertialRelativity_STE_cases`.

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